Project Development Timeline

Financial Ratio Analysis Engine — 6-Day Build Schedule

DAY 1 Database Loading

- Set up the SQLite database file **nifty100.db** to host all project data.
- Ingest raw source data — Profit & Loss, Balance Sheet, Cash Flow, Market Cap, and Analysis tables — for Nifty 100 companies.
- Validate schema, column names, and data types for each loaded table.
- Confirm row counts per table (e.g. 1,079 balance sheet rows, 1,065 cash flow rows) before proceeding.

DAY 2 Merge Engine

- Design the join strategy keyed on **company_id** and **year** via the **id_pl** reference.
- Progressively left-merge tables: Profit + Balance → + Cash Flow → + Market Cap → + Analysis.
- Track shape at each merge step to catch row loss or duplication early: (1,079, 26) → (1,065, 31) → (1,065, 37) → (1,209, 41).
- Produce the final consolidated dataset of **1,209 rows × 41 columns** ready for ratio computation.

DAY 3 Ratio Engine

- Build the pandas-based calculation module to derive ratios from the merged dataset.
- Implement core metrics: Net Profit Margin %, ROE %, ROCE %, ROA %, Interest Coverage, Asset Turnover, Free Cash Flow.
- Handle edge cases such as zero interest expense or missing early-year data, allowing NaN rather than errors.
- Preview computed ratios against a sample company (ABB) to sanity-check formulas.

DAY 4 Health Score

- Define a composite **Financial Health Score** combining profitability, efficiency, and solvency signals.
- Weight and rule-map the individual ratios (margin, ROE, ROCE, ROA, interest coverage) into a single rating.
- Validate the score against known trends — e.g. ABB's score rising from 3 to 4 alongside improving ROCE/ROA.
- Add the score as a column in the outgoing ratios table.

DAY 5 Testing

- Run the full pipeline end-to-end and check for execution errors across all companies and years.
- Spot-check outputs via SQL queries (e.g. **SELECT * FROM financial_ratios LIMIT 10**) against source data.
- Verify row counts match expectations at every stage (load → merge → ratio → save).
- Confirm NaN handling is correct and does not silently drop valid rows.

DAY 6 SQLite Integration

- Finalize persistence of results into the **financial_ratios** table inside nifty100.db.
- Confirm successful save of all **1,209 rows** with the full 9-column ratio schema.
- Close database connections cleanly after each read/write operation.
- Document the final schema and hand off the database for downstream querying and reporting.